SPECTRAL SEQUENCE OF A FILTERED COMPLEX OF MODULES ON A RINGED SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof opens with the author's own parenthetical pointing to a remark for the bounded below case with finite filtrations; the parenthetical and that reference are reproduced as written and carried as a navigation pointer.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-spectral-sequence-filtered-object is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Embedding of a filtered complex into a filtered K-injective complex. Injectives, Lemma injectives-lemma-K-injective-embedding-filtration, invoked at proof lines 6843-6846. Supplies the representative on which the spectral sequence is constructed.
- Spectral sequence of a filtered complex. Homology, Section homology-section-filtered-complex, invoked at proof lines 6847-6854. Supplies the construction of the spectral sequence from the filtered complex of global sections.
- Boundedness and convergence of the spectral sequence under the trivial vanishing conditions. Homology, Lemma homology-lemma-ss-converges-trivial, invoked at proof lines 6857-6858. Supplies the convergence statement of the second half of the lemma.

Source reference. Original lemma 0BK5: sources/stacks-chow/tags.txt, source lines 14548–14548. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 012K: sources/stacks-chow/tags.txt, source lines 1402–1402. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BKI: sources/stacks-chow/tags.txt, source lines 14561–14561. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}^\bullet$ be a filtered complex of $\mathcal{O}_X$-modules. There exists a canonical spectral sequence $(E_r, d_r)_{r\geq 1}$ of bigraded $\Gamma(X, \mathcal{O}_X)$-modules with d_r of bidegree $(r, -r+1)$ and*

$$E_1^{p,q} = H^{p+q}(X, gr^p\mathcal{F}^\bullet)$$

If for every n we have

$$H^n(X, F^p\mathcal{F}^\bullet) = 0 \text{ for } p \gg 0 \quad \text{and} \quad H^n(X, F^p\mathcal{F}^\bullet) = H^n(X, \mathcal{F}^\bullet) \text{ for } p \ll 0$$

then the spectral sequence is bounded and converges to $H^*(X, \mathcal{F}^\bullet)$.

Proof. (For a proof in case the complex is a bounded below complex of modules with finite filtrations, see the remark below.) Choose an map of filtered complexes $j : \mathcal{F}^\bullet \to \mathcal{J}^\bullet$ as in Injectives, Lemma 0BKI. The spectral sequence is the spectral sequence of Homology, Section 012K associated to the filtered complex

$$\Gamma(X, \mathcal{J}^\bullet) \quad \text{with} \quad F^p\Gamma(X, \mathcal{J}^\bullet) = \Gamma(X, F^p\mathcal{J}^\bullet)$$

Since cohomology is computed by evaluating on K-injective representatives we see that the E_1 page is as stated in the lemma. The convergence and boundedness under the stated conditions follows from Homology, Lemma 0BK5. □